

Characters of the Partition Algebras

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Frobenius [Fr] determined the irreducible characters of the symmetric group by showing that they form the change of basis matrix between power symmetric functions and Schur functions. Schur [Sc1,2] later showed that this is a consequence of the fact that the symmetric group and the general linear group generate full centralizers of each other on tensor space. In his book, *The Classical Groups*, Weyl [Wy] uses this duality to study representations and invariants for the general linear, orthogonal, symplectic, and symmetric groups. In 1937, Brauer [Br] analyzed Schur-Weyl duality for the symplectic and orthogonal groups and gave a combinatorial description of their centralizer algebras on tensor space. These centralizers are now called Brauer algebras.

Only recently has Schur-Weyl duality been studied for the last of Weyl's classical groups. If $V = \mathbb{C}^r$ is the permutation representation of the symmetric group S_r , then its centralizer algebra has a combinatorial basis given by the collection of set partitions of $\{1, 2, \dots, 2n\}$. The algebra is denoted $P_n(r)$ and is called the partition algebra. Since S_r lives inside the general linear and orthogonal group (as the subgroup of permutation matrices), the partition algebra contains the Brauer algebra and the symmetric group S_n (this is a different symmetric group which acts on $\otimes^n V$ by tensor place permutations).

The partition algebra appeared independently in the work of P.P. Martin [Ma1-4] and V.F.R. Jones [Jo]. They both study the partition algebra as a generalization of the Temperley-Lieb algebra and the Potts model in statistical mechanics. The algebra appears implicitly in [Ma1] and [Ma2, Chapter 3] and explicitly in [Ma3]. Jones [Jo] explicitly lays out the Schur-Weyl duality between $P_n(r)$ and the symmetric group S_r . Martin [Ma3,4] and Martin and Woodcock [MW1,2] extensively study the structure of the

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general partition algebra $P_n(\xi)$, with parameter $\xi \in \mathbb{C}$. They show that $P_n(\xi)$ is semisimple whenever ξ is not an integer in $[0, 2n - 1]$, and they analyze the irreducible representations in both the semisimple and non-semisimple cases.

A. Ram [Ra2] uses the duality between the orthogonal group and the Brauer algebra to derive a Frobenius formula for the Brauer algebra, and he uses symmetric functions to determine a recursive Murnaghan-Nakayama rule for Brauer algebra characters. In this paper, we determine analogues of the Frobenius formula and the Murnaghan-Nakayama rule for the characters of the partition algebras. The Frobenius formulas for the symmetric group and the Brauer algebra are identities in the ring of symmetric functions. Our analog lives in the character ring of the symmetric group. Our Murnaghan-Nakayama rule contains, as a special case, the Murnaghan-Nakayama rule for the symmetric group.

The Schur-Weyl duality between the symmetric group and the general linear group and between the orthogonal group and the Brauer algebra each have quantum generalizations (see [CP, Theorem 10.2.5] and the references there). The Iwahori-Hecke algebra of type A is a q -generalization of the symmetric group, and it is in Schur-Weyl duality with the quantum general linear group on tensor space. The Birman-Murakami-Wenzl algebra is a q -generalization of the Brauer algebra, and it is in Schur-Weyl duality with the quantum orthogonal group on tensor space. Moreover, Frobenius formulas and Murnaghan-Nakayama rules for the Iwahori-Hecke algebra [Ra1] and the Birman-Murakami-Wenzl algebra [HR] have been computed. However, to our knowledge, there is no q -generalization of the partition algebra.

Summary of Results

(1) In section 1 we develop the representation theory of the partition algebra $P_n(r)$ using double centralizer theory and the representation theory of the symmetric group S_r .

(2) In section 2, we generalize the cycle type of a permutation in the symmetric group to work for elements in the partition algebra. The “type” of a basis element in $P_n(r)$ is given by an integer partition μ with $0 \leq |\mu| \leq n$. We show that the value of any character on a basis element d is determined by its value on d_μ , where μ is the type of d . These special elements d_μ are analogous to conjugacy class representatives in the symmetric group.

(3) In section 3, we derive a formula,

$$r^{n-|\mu|} f_\mu(\rho) = \sum_{\substack{\lambda \vdash r \\ |\lambda^*| \leq n}} \chi_{S_r}^\lambda(\gamma_\rho) \chi_{P_n(r)}^\lambda(d_\mu),$$

relating S_r characters and $P_n(r)$ characters. The function f_μ is a class function on the symmetric group S_r which is explicitly computed in Theorem 3.2.2. The partition algebra character $\chi_{P_n(r)}^\lambda(d_\mu)$ can then be computed using the usual inner product $\langle r^{n-|\mu|} f_\mu, \chi_{S_r}^\lambda \rangle$ on symmetric group characters. This formula is analogous to Frobenius' formula for S_n characters.

(4) In section 4, we compute an inner product (Proposition 4.2.1) in the character ring of S_r which allows us to derive a recursive formula (Theorem 4.2.2) for $\chi_{P_n(r)}^\lambda(d_\mu)$. An interesting thing about this formula is that, like the Murnaghan-Nakayama rule for symmetric group characters, the recursion works by removing border strips from the partition λ , only in this case *we remove border strips and then put them back on*.

(5) For an indeterminate x , the generic partition algebra $P_n(x)$ over $\mathbb{C}(x)$ is semisimple, and for $\xi \in \mathbb{C}$ the specialized algebra $P_n(\xi)$ is semisimple whenever ξ is not an integer in the range $[0, 2n-1]$. We show that for all but a finite number of $\xi \in \mathbb{C}$, the characters of $P_n(\xi)$ are the evaluation of the corresponding $P_n(x)$ characters at $x = \xi$. We use this fact to lift the Murnaghan-Nakayama rule from $P_n(r)$ to $P_n(x)$ in Theorem 4.2.4.

(6) Let $\Xi_{P_n(x)}$ denote the character table of the partition algebra $P_n(x)$, that is the table of characters which has rows indexed by the irreducible representations and columns indexed by the elements described in (1). Using our character formula and suitable orderings on the rows and columns, we see that

$$\Xi_{P_n(x)} = \begin{bmatrix} x\Xi_{P_{n-1}(x)} & \vdots & * \\ \cdots & & \cdots \\ 0 & \vdots & \Xi_{S_n} \end{bmatrix}, \quad (0.1)$$

where Ξ_{S_n} is the character table of S_n . Our rule allows us to efficiently compute the values for $*$, and, in the lower-right corner of the table, it specializes to the Murnaghan-Nakayama rule for S_n . We give character tables for $P_n(x)$ when $n = 2, 3, 4, 5$ in section 6.

(7) There is a natural embedding $P_{n-1}(x) \subseteq P_n(x)$ which has special properties that are exploited by Martin [Ma3,4] to construct the irreducible representations of $P_n(x)$. In section 5, we prove general results about the characters of algebras with these properties. In particular, Proposition 5.1.9 tells us that the character table $\Xi_{P_n(x)}$ must have the form shown in (0.1). The Murnaghan-Nakayama rule is still needed, however, to directly compute the entries from $*$.

The general construction in section 5 holds for the rook monoid (see [Mu1,2], [So2]), the Solomon-Iwahori algebras ([So1], [So3], [Pu], [DHP]), and the “rook” version of the Brauer and Birman-Murakami-Wenzl algebras. The construction is similar to the Jones basic construction, which is used in

[HR] to compute the characters of the Temperley-Lieb, Brauer, Birman-Murakami-Wenzl, and Okada algebras.

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1. PARTITION ALGEBRAS

In this section we give the combinatorial definition of the partition algebras and describe its irreducible representations using its duality with the symmetric group on tensor space.

1.1. Integer Partitions and Symmetric Functions

A composition λ of the positive integer n , denoted $\lambda \models n$, is a sequence of nonnegative integers $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_t)$ such that $|\lambda| = \lambda_1 + \dots + \lambda_t = n$. The composition λ is a partition, denoted $\lambda \vdash n$, if $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_t$. The length $\ell(\lambda)$ is the number of nonzero parts of λ . Given two partitions λ, μ , we say that $\mu \subseteq \lambda$ if $\mu_i \leq \lambda_i$ for all i . If $\mu \subseteq \lambda$, let λ/μ denote the skew shape given by removing μ from λ . The size of a skew shape is $|\lambda/\mu| = |\lambda| - |\mu|$. The Young (or Ferrers) diagram of λ is the left-justified array of boxes with λ_i boxes in the i th row. For example,

$$(5, 5, 3, 1) = \begin{array}{|c|c|c|c|c|} \hline \square & \square & \square & \square & \square \\ \hline \square & \square & \square & \square & \square \\ \hline \square & \square & \square & \square & \square \\ \hline \square & \square & \square & \square & \square \\ \hline \end{array} \quad (5, 5, 3, 1)/(3, 2, 1) = \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \end{array}$$

Let $m_i(\lambda)$ denote the number of parts λ_i of λ which are equal to i . We sometimes use the notation $\lambda = (1^{m_1(\lambda)}, 2^{m_2(\lambda)}, \dots)$. In this notation the partition above is given by $(1, 3, 5^2)$.

For a positive integer k define the power symmetric function in the variables $x_1, \dots, x_r$ to be

$$p_k(x_1, \dots, x_r) = x_1^k + \dots + x_r^k,$$

and for a partition μ define

$$p_\mu = p_{\mu_1} p_{\mu_2} \cdots p_{\mu_{\ell(\mu)}}.$$

For a partition $\lambda \vdash r$, a column strict tableau T of shape λ is a filling of the boxes of the Young diagram of λ with entries from $\{1, \dots, r\}$ in such

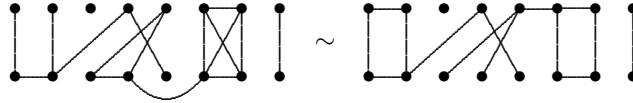
a way that the entries in the columns strictly increase when read from top to bottom and the entries in the rows weakly increase when read from left to right. For a column strict tableau T , let $m_i(T)$ denote the number of entries of T equal to i . The Schur function can be defined as

$$s_\lambda(x_1, \dots, x_r) = \sum_T x_1^{m_1(T)} x_2^{m_2(T)} \dots x_r^{m_r(T)},$$

where the sum is over all column strict tableaux of shape λ .

1.2. Set Partitions and Partition Algebras

We consider simple graphs on two rows of n -vertices, one above the other. The connected components of such a graph partition the $2n$ vertices into k disjoint subsets with $1 \leq k \leq 2n$. We say that two diagrams are equivalent if they give rise to the same partition of the $2n$ vertices. For example, the following are equivalent graphs.



We will use the term *partition diagram* (or sometimes *n-partition diagram* to indicate the number of dots) to mean the equivalence class of the given graph.

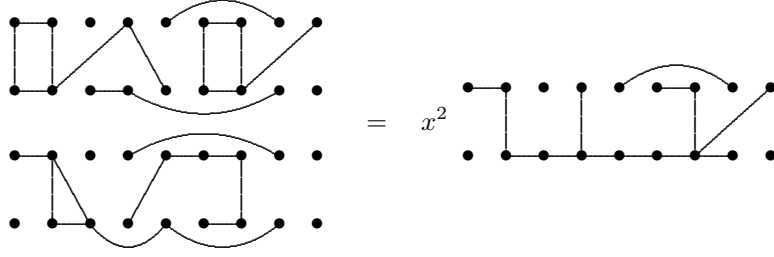
The number of n -partition diagrams which partition the $2n$ vertices into exactly k subsets is the Stirling number $S(2n, k)$, and the total number of n -partition diagrams is the Bell number (see, for example, [Mac] I.2, ex. 11)

$$B(2n) = \sum_{k=1}^{2n} S(2n, k). \quad (1.2.1)$$

Let x be an indeterminate. We multiply two n -partition diagrams d_1 and d_2 as follows:

1. Place d_1 above d_2 and identify the vertices in the bottom row of d_1 with the corresponding vertices in the top row of d_2 . This partition diagram now has a top row, a bottom row, and a middle row.
2. Let γ denote the number of subsets in this new partition diagram which contain vertices only from the middle row.
3. Let d_3 denote the n -partition diagram obtained by using only the top and bottom row (ignoring the middle row) with the imposed connections. (Note that we have removed the γ disjoint subsets that were only in the middle row).
4. The product of d_1 and d_2 is then given by $d_1 d_2 = x^\gamma d_3$.

For example,

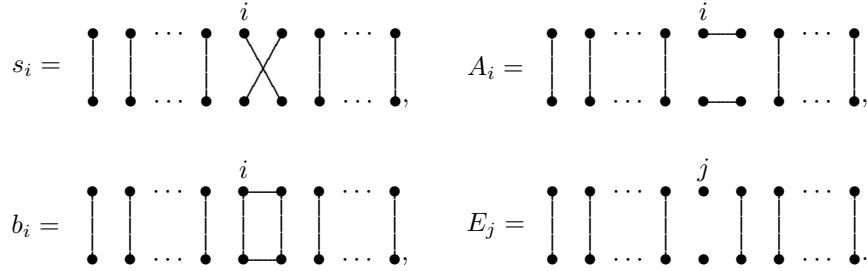


This product is associative and is independent of the graph that we choose to represent the n -partition diagram.

Let $\mathbb{C}(x)$ be the field of rational functions with complex coefficients in an indeterminate x . The *partition algebra* $P_n(x)$ is defined to be the $\mathbb{C}(x)$ -span of the n -partition diagrams. Partition diagram multiplication makes $P_n(x)$ into an associative algebra whose identity id_n is given by the partition diagram having each vertex in the top row connected to the vertex below it in the bottom row. The dimension of $P_n(x)$ is the Bell number $B(2n)$ (see (1.2.1)), and by convention $P_0(x) = \mathbb{C}(x)$.

The Brauer algebra $B_n(x)$ (see [Br]) is embedded in $P_n(x)$ as the span of the partition diagrams for which each edge is adjacent to exactly two vertices. The group algebra $\mathbb{C}(x)[S_n]$ of the symmetric group S_n is embedded in $P_n(x)$ as the span of the partition diagrams for which each edge is adjacent to exactly two vertices, one in each row.

For $1 \leq i \leq n-1$ and $1 \leq j \leq n$, let



Notice that we have the relations

$$(a) \ A_i^2 = xA_i, \quad (b) \ E_j^2 = xE_j, \quad (c) \ A_i = b_i E_i E_{i+1} b_i. \quad (1.2.2)$$

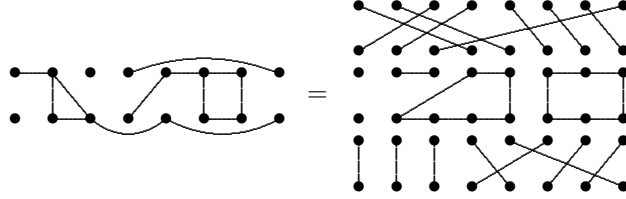
For $1 \leq i \leq n-1$ and $1 \leq j \leq n$, we define

$$a_i = \frac{1}{x} A_i \quad \text{and} \quad e_j = \frac{1}{x} E_j \quad (1.2.3)$$

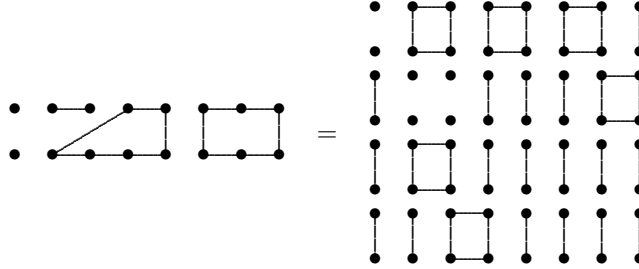
so that a_i and e_j are idempotent.

The element s_i corresponds to the simple transposition $s_i = (i, i+1) \in S_n$, and the elements of the set $\{s_i \mid 1 \leq i \leq n-1\}$ generate $\mathbb{C}(x)[S_n]$. The elements $\{s_i, A_i \mid 1 \leq i \leq n-1\}$ generate the Brauer algebra $B_n(x)$.

The elements $\{s_i, b_i, E_j \mid 1 \leq i \leq n-1, 1 \leq j \leq n\}$ generate all of $P_n(x)$. To see this, think of a diagram d as living within the rectangle which has as its corners the vertices $1, n, n+1$ and $2n$. Following [Jo], we say that a partition diagram d is *planar* if it is possible to draw a collection of non-intersecting closed curves inside the rectangle of d which partition the vertices of d into its equivalence classes. It is then possible to write $d = \sigma p \tau$, where $\sigma, \tau \in S_n$ and p is a planar partition diagram (see [Jo], Lemma 2, for the obvious proof of this fact). For example, the diagram d_2 , above, decomposes as



Now, it is easy to see that each planar partition diagram can be written as a product of b_i , $0 \leq i \leq n-1$, and E_j , $0 \leq j \leq n$. The planar diagram in our example can be written as



It is well-known and easy to prove by induction that the number of planar partition diagrams is the Catalan number $\frac{1}{2n+1} \binom{4n}{2n}$. Moreover, the product of two planar partition diagrams is planar, so the span of these diagrams is a subalgebra $K(2n, x)$.

For each complex number $\xi \in \mathbb{C}$, one defines a partition algebra $P_n(\xi)$ over $\mathbb{C}$ as the linear span of n -partition diagrams where the multiplication is as given above except with x replaced by ξ . The following theorem can be proved using the same argument that Wenzl [Wz] uses for the Brauer algebra, only with the dimension of the irreducible S_r -module $\dim(S^\lambda) =$

$r!/h_\lambda$ replacing the El Samra-King polynomials. Here h_λ is the product of the hooks in λ (see [Mac]).

THEOREM 1.2.4. (Martin and Saleur [MS]) *For each integer $n \geq 0$, $P_n(x)$ is semisimple over $\mathbb{C}(x)$, and $P_n(\xi)$ is semisimple over $\mathbb{C}$ whenever ξ is not an integer in the range $[0, 2n - 1]$.*

1.3. Schur-Weyl Duality

Let $V = \mathbb{C}^r$ with standard basis $v_1, v_2, \dots, v_r$ be the permutation module for the symmetric group S_r . Thus,

$$\sigma(v_i) = v_{\sigma(i)}, \quad \text{for } \sigma \in S_r \text{ and } 1 \leq i \leq r. \quad (1.3.1)$$

For each positive integer n , the tensor product space $\otimes^n V$ is a module for S_r with a standard basis given by $v_{i_1} \otimes v_{i_2} \otimes \dots \otimes v_{i_n}$, where $1 \leq i_j \leq r$. The action of $\sigma \in S_r$ on a basis vector is given by

$$\sigma(v_{i_1} \otimes v_{i_2} \otimes \dots \otimes v_{i_n}) = v_{\sigma(i_1)} \otimes v_{\sigma(i_2)} \otimes \dots \otimes v_{\sigma(i_n)}. \quad (1.3.2)$$

Number the vertices of an n -partition diagram $1, 2, \dots, n$ from left to right in the top row and $n+1, n+2, \dots, 2n$ from left to right in the bottom row. For each n -partition diagram d and each integer sequence $i_1, \dots, i_{2n}$ with $1 \leq i_k \leq r$, define

$$\delta(d)_{i_{n+1}, \dots, i_{2n}}^{i_1, \dots, i_n} = \begin{cases} 1, & \text{if } i_j = i_k \text{ whenever vertex } j \text{ is} \\ & \text{connected to vertex } k \text{ in } d, \\ 0, & \text{otherwise.} \end{cases} \quad (1.3.3)$$

Define an action of a partition diagram $d \in P_n(r)$ on $\otimes^n V$ by defining it on the standard basis by

$$(v_{i_1} \otimes v_{i_2} \otimes \dots \otimes v_{i_n})d = \sum_{i_{n+1}, \dots, i_{2n}} \delta(d)_{i_{n+1}, \dots, i_{2n}}^{i_1, \dots, i_n} v_{i_{n+1}} \otimes v_{i_{n+2}} \otimes \dots \otimes v_{i_{2n}}. \quad (1.3.4)$$

THEOREM 1.3.5. (Jones [Jo]) *S_r and $P_n(r)$ generate full centralizers of each other in $\text{End}(\otimes^n V)$. In particular, for $r \geq 2n$,*

- (a) $P_n(r)$ generates $\text{End}_{S_r}(\otimes^n V)$, and when $r \geq 2n$, $P_n(r) \cong \text{End}_{S_r}(\otimes^n V)$.
- (b) S_r generates $\text{End}_{P_n(r)}(\otimes^n V)$.

Proof. We have $A \in \text{End}_{S_r}(\otimes^n V)$ if and only if $\sigma^{-1}A\sigma = A$ as operators on $\otimes^n V$ for each $\sigma \in S_r$. Let $A = [A_{i_{n+1}, i_{n+2}, \dots, i_{2n}}^{i_1, i_2, \dots, i_n}]$. Since V is the permutation representation for S_r , $\sigma^{-1}A\sigma$ has matrix $[A_{\sigma(i_{n+1}), \sigma(i_{n+2}), \dots, \sigma(i_{2n})}^{\sigma(i_1), \sigma(i_2), \dots, \sigma(i_n)}]$.

Hence the entries of the matrix of A are equal for all coordinates in the orbit of σ . If we collect the $i_1, i_2, \dots, i_{2n}$ according to those that have an equal value, we form a partition of $\{1, \dots, 2n\}$ into subsets, and the indices in each subset will always have equal value in an orbit of σ .

Given an equivalence relation $\sim$ which partitions the set $\{1, 2, \dots, 2n\}$ into at most r subsets, define $T_\sim \in \text{End}(\otimes^n V)$ by

$$[(T_\sim)_{i_{n+1}, i_{n+2}, \dots, i_{2n}}^{i_1, i_2, \dots, i_n}] = \begin{cases} 1, & \text{if } i_j = i_k \text{ whenever } j \sim k \\ 0, & \text{otherwise.} \end{cases}$$

Then, the set $T_\sim$, as $\sim$ ranges over all partitions of $\{1, 2, \dots, 2n\}$ into at most r subsets, forms a basis of $\text{End}_{S_r}(\otimes^n V)$. Moreover, this is exactly the matrix of the corresponding partition diagram (see 1.3.3). When $r \geq 2n$, all equivalence relations $\sim$ occur, and the representation of $P_n(r)$ on $\text{End}(\otimes^n V)$ is faithful. Part (b) follows since these centralizer algebras are semisimple and because $P_n(r)$ is the centralizer of S_r . ■

1.4. Irreducible Representations of $P_n(r)$

The irreducible representations of S_r are indexed by the partitions $\lambda \vdash r$, and we denote them by S^λ . The permutation module V decomposes into irreducibles as $V \cong S^{(r)} \oplus S^{(r-1,1)}$, where $S^{(r)}$ is the trivial module for S_r . The rules for Kronecker products of these special symmetric group modules are known (for example they can be derived from [Mac], I.7, ex. 23(d)). For $\lambda \vdash r$, the decomposition of $S^\lambda \otimes V$ into irreducible S_r modules is

$$S^\lambda \otimes V \cong \bigoplus_{\mu=(\lambda^-)^+} V^\mu, \quad (1.4.1)$$

where the sum is over all $\mu \vdash r$ such that μ is obtained by removing a box from λ to get a partition λ^- and then adding the box back to λ^- to get μ . The Bratteli diagram for $r = 6$ is shown in Figure 1.

The partition algebra $P_{n-1}(x)$ is embedded in $P_n(x)$ by adding a horizontal edge connecting the n th and the $2n$ th dots. The lines in the Bratteli diagram indicate the branching rules under the restriction from $P_n(r)$ to $P_{n-1}(r)$. Martin [Ma4] shows that the restriction rules are the same for $P_n(x)$ and, in the semisimple cases, $P_n(\xi)$.

From double centralizer theory (see, for example [CR], §3D), we have the following remarkable consequences of Theorem 1.3.5:

(1) The irreducible representations of $P_n(r)$ are indexed by the same set that indexes the irreducible S_r -modules in $\otimes^n V$. Using (1.4.1), we see that this is the set

$$\widehat{P_n(r)} = \{ \lambda \vdash r \mid |\lambda^*| \leq n \}, \quad \text{where } \lambda^* = \lambda/(\lambda_1). \quad (1.4.2)$$

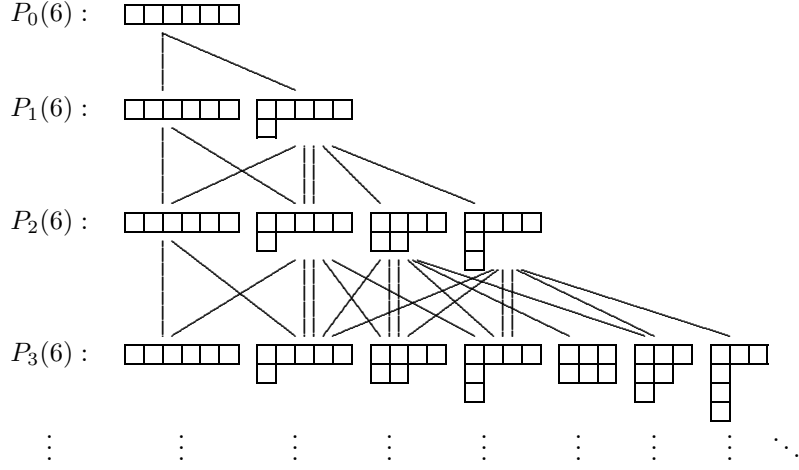
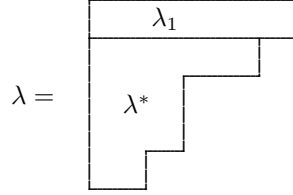


FIG. 1. The Bratteli diagram for $P_n(6)$

Here λ^* is the partition obtained from λ by removing its first part λ_1 , i.e.,



(2) Denote by M^λ the irreducible representation of $P_n(r)$ indexed by $\lambda \in \widehat{P_n(r)}$. The dimension of M^λ equals the multiplicity of S^λ in $\otimes^n V$ and thus is the number of paths from the top of the Bratteli partition diagram to λ . In row $n = 3$ of Figure 1, these dimensions are 5, 10, 6, 6, 1, 2, 1 (reading left to right). Moreover, $5^2 + 10^2 + 6^2 + 6^2 + 1^2 + 2^2 + 1 = 203$ which is the Bell number $B(6)$.

(3) The decomposition of $\otimes^n V$ as a bimodule for $S_r \times P_n(r)$ is

$$\otimes^n V \cong \bigoplus_{\lambda \in \widehat{P_n(r)}} S^\lambda \otimes M^\lambda, \quad (1.4.3)$$

and, when $r \geq 2n$, $\{M^\lambda \mid \lambda \in \widehat{P_n(r)}\}$ is a complete set of irreducible $P_n(r)$ -modules.

Remark 1.4.4. We have two symmetric groups acting on $\otimes^n V$. The group $S_r \subseteq GL(r, \mathbb{C})$ acts on the left by the tensor product of its permutation representation, and the group $S_n \subseteq P_n(r)$ acts on the right by tensor place permutations.

Remark 1.4.5. In the classical case [Sc1,2], $V = \mathbb{C}^r$ is the standard representation of $GL(r, \mathbb{C})$ where $g \in GL(r, \mathbb{C})$ acts by matrix multiplication on the left. Restriction from $GL(r, \mathbb{C})$ to S_r gives the permutation representation (1.3.1) of S_r . The centralizer algebra is $\mathbb{C}[S_n] \subseteq P_n(r)$ with the action given by (1.3.4). When $r \geq n$,

$$\otimes^n V \cong \bigoplus_{\lambda \vdash n} V^\lambda \otimes S^\lambda, \quad (1.4.6)$$

where V^λ is a $GL(r, \mathbb{C})$ -module of highest weight λ and S^λ is an irreducible S_n -module.

1.5. Irreducible Representations of $P_n(x)$

When $r \geq 2n$, the index set $\widehat{P_n(r)}$ defined in (1.3.8) is in bijection with the set

$$\widehat{P_n} = \widehat{P_n(2n)} = \{\lambda \vdash 2n \mid |\lambda^*| \leq n\}. \quad (1.5.1)$$

If $\lambda \in \widehat{P_n(r)}$, the bijection is given by subtracting $r - 2n$ from λ_1 . Since $r \geq 2n$ and $|\lambda^*| \leq n$, we see that $\lambda_1 - \lambda_2 \geq (r - n) - n = r - 2n$, and thus this subtraction will always give a partition of $2n$.

Martin [Ma3] shows that, in the semisimple cases, the irreducible representations of $P_n(x)$ and $P_n(\xi)$ are indexed by the partitions $\mu \vdash k, 0 \leq k \leq n$. These partitions are also in bijection with $\widehat{P_n}$. Indeed, they are *exactly* the partitions λ^* in (1.5.1) and (1.4.2). We get from Martin's partitions to those in $\widehat{P_n}$ by adding a top row with $2n - k$ boxes. The partitions in $\widehat{P_n}$ are better suited for the recursive algorithm that we derive in section 4.

For $\lambda \in \widehat{P_n}$, let $\chi_{P_n(\xi)}^\lambda$ denote the irreducible character of $P_n(\xi)$ corresponding to M^λ . The proof of the next proposition is identical to that of Corollary 2.4 in [Ra2].

PROPOSITION 1.5.2. *For all but a finite number of $\xi \in \mathbb{C}$, the character $\chi_{P_n(\xi)}^\lambda$ of $P_n(\xi)$ equals the character $\chi_{P_n(x)}^\lambda$ of $P_n(x)$ evaluated at $x = \xi$.*

2. CONJUGACY CLASS ANALOGS

In this section, we divide the partition algebra diagrams into classes on which characters are constant. These are analogs of the conjugacy classes, determined by cycle type, in the symmetric group.

2.1. Standard Elements

Let $\gamma_1 = 1$, for $t > 1$ let $\gamma_t = s_{t-1}s_{t-2}\cdots s_1$, and let E_1 be the 1-partition diagram with no edges. Thus,

$$\gamma_t = \begin{array}{c} 1 \quad 2 \quad \cdots \quad t \\ \bullet \quad \bullet \quad \cdots \quad \bullet \\ \bullet \quad \bullet \quad \cdots \quad \bullet \end{array} \quad \text{and} \quad E_1 = \begin{array}{c} \bullet \\ \bullet \end{array} \quad (2.1.1)$$

If d_1 and d_2 are n_1 and n_2 -partition diagrams respectively, then $d_1 \otimes d_2$ is the $(n_1 + n_2)$ -partition diagram obtained by placing d_2 to the right of d_1 . For $\mu = (\mu_1, \dots, \mu_\ell) \models k$, define

$$\gamma_\mu = \gamma_{\mu_1} \otimes \cdots \otimes \gamma_{\mu_\ell} \in S_k,$$

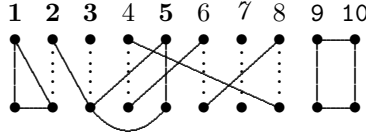
and for a composition μ with $0 \leq |\mu| \leq n$, define

$$d_\mu = \gamma_\mu \otimes E_1^{\otimes(n-|\mu|)} = \gamma_\mu \otimes \underbrace{E_1 \otimes \cdots \otimes E_1}_{n-|\mu| \text{ times}} \in P_n(x), \quad (2.1.2)$$

The cycle type of a permutation $\sigma \in S_n$ is the partition $\rho(\sigma) \vdash n$ given by the lengths of the disjoint cycles in σ . Each $\sigma \in S_n$ is conjugate to $\gamma_{\rho(\sigma)}$, and $\{\gamma_\rho \mid \rho \vdash n\}$ is a complete set of S_n -conjugacy class representatives.

2.2. “Cycle Type” in the Partition Algebras.

Let $d \in P_n(x)$ be a partition diagram. The vertices of d are arranged in n columns of size 2. Connect each vertex in d to the other vertex in its column with a dotted line. The connected components of this new diagram partition the vertices into disjoint subsets, called *blocks*. Vertices in the same column are always in the same block, and there are no connections between disjoint blocks. The following diagram consists of four blocks. The first is on columns 1, 2, 3, 5, the second on columns 4, 6, 8, the third on column 7, and the fourth on columns 9, 10.



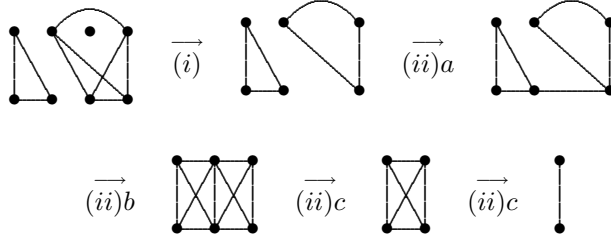
A vertex in d is *isolated* if it is incident to no edges, and an edge in d is *horizontal* if it connects vertices in the same row. If a vertex is adjacent to

two vertices in the opposite row, then those two vertices are adjacent by a horizontal edge. Thus, if d has no isolated vertices and no horizontal edges, then $d \in S_n$. Furthermore, a single block in an S_n diagram is a cycle.

We assign a nonnegative integer to each block, called the *block type*, using the following algorithm:

1. While the block is not E_1 or a permutation diagram, do the following:
 - (i) If the block has an isolated vertex v , then remove the column containing v from the block (removing the edges incident to the other vertex in same column as v).
 - (ii) If the block has a horizontal edge, then
 - a. connect the corresponding vertices in the opposite row by an edge
 - b. add any new edges that are now implied by transitivity
 - c. remove one of these two columns from the diagram.
2. If the remaining block is E_1 , then it has block type 0. Otherwise, the remaining block is a permutation diagram and so it is a cycle. The block type in this case is the length of this cycle, i.e., the number of columns remaining in the block.

For example, we apply the algorithm to the first block in the example above:



and this block has type 1.

The sequence of block types of d forms a partition $\mu = (0^{m_0}, 1^{m_1}, \dots)$, where m_i is the number of blocks of type i . Note that

$$n - |\mu| - m_0 = (\text{the number of columns removed in the algorithm}).$$

In our example, the blocks have type 1, 3, 0, and 1, respectively, and so the block type of the diagram is $(0, 1^2, 3)$. Here $n = 10$ and $10 - 5 - 1 = 4$. Three columns were removed from the first block in the illustration above, and we remove 1 column when determining the type of the last block.

If $d, d' \in P_n(x)$, then we say that d' is conjugate to d if there exists $\pi \in S_n$ such that $d' = \pi d \pi^{-1}$. The following properties are easy to verify:

1. If d is a partition diagram, then $\pi d \pi^{-1}$ is the partition diagram given by rearranging the vertices of d , in both the top and bottom row, according to π .
2. If χ is any character of $P_n(x)$, then $\chi(\pi d \pi^{-1}) = \chi(d)$.
3. $d_1 \otimes d_2$ is conjugate to $d_2 \otimes d_1$.
4. $\pi^{-1} d \pi$ has the same block type as d .
5. If $d \in P_n(x)$ has blocks of size $k_1 \geq k_2 \geq \dots \geq k_\ell$, then there exists $\pi \in S_n$ so that

$$\pi^{-1} d \pi = b_1 \otimes b_2 \otimes \dots \otimes b_\ell,$$

where each b_i consists of a single block of size k_i in $P_{k_i}(x)$. Simply choose π to rearrange the vertices so that those in the largest block come first, those in the next largest block come next, and so on (breaking ties arbitrarily).

6. Each cycle of length t is conjugate to γ_t .

PROPOSITION 2.2.1. *If d is a partition diagram with block type $\mu = (0^{m_0}, 1^{m_1}, \dots)$ and χ is any character of $P_n(x)$, then*

$$\chi(d) = x^{-(n-|\mu|-m_0)} \chi(d_\mu),$$

where $d_\mu = \gamma_\mu \otimes E_1^{\otimes(n-|\mu|)}$ is a standard diagram defined in (2.1.2).

Proof. From 1-5 above, we see that we can work on the individual blocks of d independently, and thus we assume that d consists of a single block. If there is only one column in d , then we are done, since the two possible partition diagrams in this case are of the form (2.1.2). Thus, we assume that d has more than 1 column.

(i) If d has an isolated vertex at position i in the top row, then the vertex below it is not isolated, since d is a block with more than one column. Thus

$$E_i d = x d \quad \text{and} \quad d E_i = d'$$

where d' is the same diagram as d only the edges incident to the i th vertex in the bottom row have been removed. By permuting the i th column to the end, d' is conjugate to $d'' \otimes E_1$ where d'' is the diagram obtained from d by removing the i th column. Thus,

$$\chi(d) = \frac{1}{x} \chi(E_i d) = \frac{1}{x} \chi(d E_i) = \frac{1}{x} \chi(d'' \otimes E_1).$$

The symmetric argument is used when the isolated dot is in the bottom row.

(ii) If d has a horizontal edge in the top row connecting vertices i and j , define

$$b = \begin{array}{c} \text{ } \\ \text{ } \end{array} \begin{array}{ccccccc} & & i & & j & & \\ \bullet & \cdots & \bullet & \cdots & \bullet & \cdots & \bullet \\ \vdots & & \vdots & & \vdots & & \vdots \\ \bullet & \cdots & \bullet & \cdots & \bullet & \cdots & \bullet \end{array}$$

Then

$$bd = d \quad \text{and} \quad db = d'$$

where d' has an isolated vertex in the bottom row of the j th column. Now case (i) applies to get $\chi(d) = \frac{1}{x}\chi(d' \otimes E_1)$ where d' is obtained from d by connecting the i th and j th vertices in the bottom row, connecting any new edges implied by transitivity, and then removing the j th column. Again, the symmetric argument is used for horizontal edges in the bottom row.

Each application of (i) or (ii) removes one column from the diagram and adds E_1 at the end. Eventually we reach a permutation diagram (no isolated vertices and no horizontal edges), which is conjugate to a cycle γ_t , or we reach E_1 . The argument is the same for $P_n(\xi)$, $\xi \in \mathbb{C}$, $\xi \neq 0$, only with ξ replacing x . ■

3. THE FROBENIUS FORMULA

For a partition $\lambda \vdash r$, let $\chi_{S_r}^\lambda$ denote the irreducible character of the symmetric group S_r corresponding to S^λ . If $\lambda \vdash r$ and $|\lambda^*| \leq n$ (see (1.4.2)), then let $\chi_{P_n(r)}^\lambda$ denote the irreducible character of the partition algebra $P_n(r)$ corresponding to M^λ .

3.1. The Frobenius Formula for the Symmetric Group

As in sections 1.3 and 1.4, let $V = \mathbb{C}^r$ be the natural representation of $GL(r, \mathbb{C})$, and let $\otimes^n V$ be its n -fold tensor product. Let S_n act on $\otimes^n V$ by the action defined in (1.3.4). Define the bitrace of $g \times \tau \in GL(r, \mathbb{C}) \times S_n$ on $\otimes^n V$ to be

$$btr(g, \tau) = \sum_{i_1, \dots, i_n} g(v_{i_1} \otimes \cdots \otimes v_{i_n}) \tau|_{v_{i_1} \otimes \cdots \otimes v_{i_n}} \quad (3.1.1)$$

where $g(v_{i_1} \otimes \cdots \otimes v_{i_n}) \tau|_{v_{i_1} \otimes \cdots \otimes v_{i_n}}$ denotes the coefficient of $v_{i_1} \otimes \cdots \otimes v_{i_n}$ in the expansion of $g(v_{i_1} \otimes \cdots \otimes v_{i_n}) \tau$. Since the actions of $GL(r, \mathbb{C})$ and S_n commute, the bitrace is a trace in both components, and thus it is sufficient to compute the bitrace on conjugacy class representatives.

For $g \in GL(r, \mathbb{C})$ with eigenvalues $x_1, \dots, x_r$ and $\gamma_\mu \in S_n$ with $\mu \vdash n$ as in (2.1.2), Schur [Sc1,2] proved that the bitrace is given by the power

symmetric function,

$$btr(g, \gamma_\mu) = p_\mu(x_1, \dots, x_r). \quad (3.1.2)$$

The proof of (3.1.2) is a straight-forward computation (see [Ra2]). The irreducible character of $GL(r, \mathbb{C})$ corresponding to the highest weight module V^λ is the Schur function s_λ , so by computing the bitrace on both sides of (1.4.6) we obtain Frobenius' formula

$$p_\mu(x_1, \dots, x_r) = \sum_{\lambda \vdash n} s_\lambda(x_1, \dots, x_r) \chi_{S_r}^\lambda(\gamma_\mu). \quad (3.1.3)$$

3.2. A Frobenius Formula for the Partition Algebras

Let $\rho \vdash r$ and $\mu \models m$ with $0 \leq m \leq n$. Then let $\sigma \in S_r$, and let $d_\mu \in P_n(r)$ be defined as in (2.1.2). Since the actions of S_r and $P_n(r)$ commute on $\otimes^n V$, we define the bitrace of $btr(\sigma, d_\mu)$ exactly as in (3.1.1) only with σ replacing g and d_μ replacing γ_μ . Then, computing the trace on the right-hand side of (1.4.3) gives

$$btr(\sigma, d_\mu) = \sum_{\lambda \in \overline{P_n(r)}} \chi_{S_r}^\lambda(\sigma) \chi_{P_n(r)}^\lambda(d_\mu) \quad (3.2.1)$$

We now compute $btr(\sigma, d_\mu)$ directly on $\otimes^n V$.

THEOREM 3.2.2. *Let $\sigma \in S_r$ and $\mu = (\mu_1, \dots, \mu_\ell)$ be a composition with $0 \leq |\mu| \leq n$. The bitrace of $\sigma \times d_\mu \in S_r \times P_n(r)$ on $\otimes^n V$ is given by*

$$btr(\sigma, d_\mu) = r^{n-|\mu|} f_\mu(\sigma) = r^{n-|\mu|} f_{\mu_1}(\sigma) f_{\mu_2}(\sigma) \cdots f_{\mu_\ell}(\sigma)$$

where, if $\rho = (\rho_1, \rho_2, \dots)$ is the cycle type of σ and k is a positive integer, then $f_k(\sigma)$ is computed by any of the following equivalent formulas:

- (a) $f_k(\sigma) = \sum_{i=1}^{\ell(\rho)} p_k(1, \omega_{\rho_i}, \omega_{\rho_i}^2, \dots, \omega_{\rho_i}^{\rho_i-1}),$
- (b) $f_k(\sigma) = \sum_{d|k} d \cdot m_d(\rho),$ (summing over the divisors d of k),
- (c) $f_k(\sigma) = \chi_V(\sigma^k) = (\text{the number of fixed points of } \sigma^k),$

such that $\omega_j = e^{2\pi i/j}$ is a primitive j th root of unity, $m_d(\rho)$ is the number of parts of ρ equal to d , and χ_V is the character of the permutation representation V of S_r .

Proof. If $\sigma \in S_r$, d_1 is an n_1 -partition diagram, and d_2 is an n_2 -partition diagram, then $btr(\sigma, d_1 \otimes d_2) = btr(\sigma, d_1) btr(\sigma, d_2)$ where $\sigma \times d_1$ acts on the first $\otimes^{n_1} V$ and $\sigma \times d_2$ acts on the last $\otimes^{n_2} V$. It follows from (2.1.2) that

$$btr(\sigma, d_\mu) = btr(\sigma, \gamma_{\mu_1}) btr(\sigma, \gamma_{\mu_2}) \cdots btr(\sigma, \gamma_{\mu_\ell}) btr(\sigma, E_1)^{n-|\mu|}.$$

We compute the value of $btr(\sigma, E_1)$ on V directly. From (1.3.4), we see that $v_i E_1 = \sum_{j=1}^r v_j$, and thus

$$btr(\sigma, E_1) = \sum_{i=1}^r \sigma v_i E_1|_{v_i} = \sum_{i=1}^r \sum_{j=1}^r \sigma v_j|_{v_i} = \sum_{i=1}^r \sum_{j=1}^r v_{\sigma(j)}|_{v_i} = \sum_{i=1}^r 1 = r.$$

Define $f_k(\sigma) = btr(\sigma, \gamma_k)$, and we have proved

$$f_\mu(\sigma) = r^{n-|\mu|} f_{\mu_1}(\sigma) \cdots f_{\mu_\ell}(\sigma).$$

It remains to compute $f_k(\sigma)$.

(a) Using Schur's result (3.1.2) restricted to $S_r \subseteq GL(r, \mathbb{C})$, we have

$$btr(\gamma_\rho, \gamma_k) = p_k(x_1, \dots, x_r),$$

where $x_1, \dots, x_r$ are the eigenvalues of γ_ρ . The eigenvalues of a j -cycle in S_r are the j th roots of unity $1, \omega_j, \dots, \omega_j^{j-1}$, and $\sigma \in S_r$ is a disjoint product of cycles. Thus, the eigenvalues of $\sigma \in S_r$ are the union of the j th roots of unity as j ranges over the lengths of the cycles in σ . Part (a) then follows from the fact that $p_k(x_1, \dots, x_a, y_1, \dots, y_b) = p_k(x_1, \dots, x_a) + p_k(y_1, \dots, y_b)$.

(b) If $\omega_j = e^{2\pi i/j}$, then

$$\begin{aligned} p_k(1, \omega_j, \dots, \omega_j^{j-1}) &= 1 + \omega_j^k + \omega_j^{2k} + \cdots + \omega_j^{k(j-1)} \\ &= \frac{\omega_j^{kj} - 1}{\omega_j^k - 1} \\ &= \begin{cases} j, & \text{if } j \text{ divides } k, \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

Thus, the only nonzero summands which appear in (a) are when ρ_i divides k , and in that case the value is ρ_i . We can then sum the divisors d of k , and they appear with multiplicity $m_d(\rho)$.

(c) The character value $\chi_V(\sigma^k)$ is the number of fixed points of σ^k , and the fixed points of σ^k are the elements of the d -cycles of σ where d divides k . Part (c) then follows from from (b). ■

Putting (3.2.1) and Theorem 3.2.2 together gives our Frobenius formula for the characters of $P_n(r)$. For a composition μ with $0 \leq |\mu| \leq n$, we have

$$r^{n-|\mu|} f_\mu(\sigma) = \sum_{\lambda \in \widehat{P_n(r)}} \chi_{P_n(r)}^\lambda(d_\mu) \chi_{S_r}^\lambda(\sigma), \quad \text{for all } \sigma \in S_r. \quad (3.2.3)$$

For each composition μ , the function f_μ is a class function on S_r . Let $R(S_r)$ denote the $\mathbb{C}$ -vector space generated by the class functions of S_r . The irreducible characters of S_r form a basis of $R(S_r)$, and the Frobenius formula (3.2.3) says that the character $\chi_{P_n(r)}^\lambda(d_\mu)$ is the coefficient of $\chi_{S_r}^\lambda$ when $r^{n-|\mu|}f_\mu$ is expanded in terms of the irreducible S_r characters.

The module $R(S_r)$ carries a scalar product $\langle, \rangle$ which is defined by

$$\langle g, h \rangle = \sum_{\rho \vdash r} \frac{1}{z_\rho} g(\gamma_\rho) h(\gamma_\rho), \quad \text{for } g, h \in R(S_r), \quad (3.2.4)$$

where $z_\rho = \prod_{i=1}^{\ell(\rho)} i^{m_i(\rho)} m_i(\rho)!$ is the size of the centralizer of γ_ρ in S_r . With respect to this product the irreducible characters of S_r are orthonormal, and so we have the following corollary of our Frobenius formula.

COROLLARY 3.2.5. *Let μ be a composition with $0 \leq |\mu| \leq n$, and let $\lambda \vdash r$ with $|\lambda^*| \leq n$. Then*

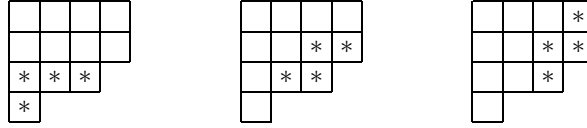
$$\chi_{P_n(r)}^\lambda(d_\mu) = \langle r^{n-|\mu|} f_\mu, \chi_{S_r}^\lambda \rangle = r^{n-|\mu|} \langle f_\mu, \chi_{S_r}^\lambda \rangle.$$

4. THE MURNAGHAN-NAKAYAMA RULE

The Murnaghan-Nakayama rule is a recursive rule for computing characters of the symmetric group. It is derived by expanding polynomials in the ring of symmetric functions. In our work here, we must derive the rule by working in the character ring of the symmetric group.

4.1. The Murnaghan-Nakayama Rule for the Symmetric Group

Two boxes in a skew shape λ/μ are adjacent if they share a common edge, and λ/μ is connected if you can travel from any box to any other via a path of adjacent boxes. A skew shape λ/μ is a *border strip* if it is connected and it does not contain any 2×2 blocks of boxes. The following figures illustrate the three border strips of size 4 in $\lambda = (4, 4, 3, 1)$.



The height of a border strip is given by

$$ht(\lambda/\mu) = \#(\text{rows occupied by } \lambda/\mu) - 1.$$

The heights of the border strips in the figures above are 1, 1, and 2, respectively. For a proof of the next theorem, see [Mac] I.7, ex. 5.

THEOREM 4.1.1. (The Murnaghan-Nakayama Rule for S_n) *Let $\lambda \vdash n$, $\mu \models n$, and let $\bar{\mu}$ be obtained from μ by removing a part of size k . Then the irreducible character $\chi_{S_n}^\lambda(\gamma_\mu)$ is given by*

$$\chi_{S_n}^\lambda(\gamma_\mu) = \sum_{\lambda^{-k} \subseteq \lambda} (-1)^{ht(\lambda/\lambda^{-k})} \chi_{S_{n-k}}^{\lambda^{-k}}(\gamma_{\bar{\mu}}),$$

where sum is over all partitions $\lambda^{-k} \subseteq \lambda$ such that λ/λ^{-k} is a border strip of size k .

4.2. A Murnaghan-Nakayama Rule for the Partition Algebras

If $\chi_{S_r}^\delta$ is a character of S_r , and k is a positive integer, then $f_k \chi_{S_r}^\delta$ is the class function which takes the value $f_k(\sigma) \chi_{S_r}^\delta(\sigma)$ on $\sigma \in S_r$. The next proposition gives the expansion of $f_k(\sigma) \chi_{S_r}^\delta(\sigma)$ in terms of the basis of irreducible characters. It is the analogue of the symmetric function identities [Mac], I.3 ex. 11(2), for the symmetric group and [Ra2], Theorem 6.8, for the Brauer algebra.

PROPOSITION 4.2.1. *Let $\delta, \lambda \vdash r$. Using the inner product $\langle, \rangle$ on S_r characters (3.2.4), we have*

$$\langle f_k \chi_{S_r}^\delta, \chi_{S_r}^\lambda \rangle = \sum_{d|k} \sum_{\substack{\lambda^{-d} \subseteq \lambda \\ \lambda^{-d} \subseteq \delta}} (-1)^{ht(\lambda/\lambda^{-d})} (-1)^{ht(\delta/\lambda^{-d})}$$

where the outer sum is over all divisors d of k and the inner the sum is over all partitions $\lambda^{-d} \vdash (r-d)$ such that $\lambda^{-d} \subseteq \lambda$, $\lambda^{-d} \subseteq \delta$ and λ/λ^{-d} and δ/λ^{-d} are border strips of size d . If no such λ^{-d} exists, then the inner product is zero.

Proof. By (3.2.4) and Theorem 3.2.2, we have

$$\begin{aligned} \langle f_k \chi_{S_r}^\delta, \chi_{S_r}^\lambda \rangle &= \sum_{\rho \vdash r} \frac{1}{z_\rho} f_k(\rho) \chi_{S_r}^\delta(\gamma_\rho) \chi_{S_r}^\lambda(\gamma_\rho) \\ &= \sum_{d|k} \sum_{\rho \vdash r} \frac{d m_d(\rho)}{z_\rho} \chi_{S_r}^\delta(\gamma_\rho) \chi_{S_r}^\lambda(\gamma_\rho). \end{aligned}$$

The nonzero terms of the inner sum occur when $\rho \vdash r$ has at least one part of size d . These partitions are in bijection with partitions of the form $\alpha + (d)$, where $\alpha \vdash (n-d)$ and $\alpha + (d)$ is the partition of r obtained by adding a part of size d to α . Moreover, for partitions of this form, we have

$z_{\alpha+(d)} = d m_d(\alpha + (d)) z_\alpha$ (see (3.2.4)), which implies that

$$\frac{d m_d(\alpha + (d))}{z_{\alpha+(d)}} = \frac{1}{z_\alpha}.$$

Thus,

$$\langle f_k \chi_{S_r}^\delta, \chi_{S_r}^\lambda \rangle = \sum_{d|k} \sum_{\alpha \vdash (r-d)} \frac{1}{z_\alpha} \chi_{S_r}^\delta(\gamma_{\alpha+(d)}) \chi_{S_r}^\lambda(\gamma_{\alpha+(d)}).$$

We now apply the Murnaghan-Nakayama rule for S_r (Theorem 4.1.1) to remove a part of size d from $\chi_{S_r}^\delta(\gamma_{\alpha+(d)})$ and $\chi_{S_r}^\lambda(\gamma_{\alpha+(d)})$:

$$\begin{aligned} \langle f_k \chi_{S_r}^\delta, \chi_{S_r}^\lambda \rangle &= \\ &= \sum_{d|k} \sum_{\alpha \vdash (r-d)} \frac{1}{z_\alpha} \sum_{\delta^{-d} \subseteq \delta} \sum_{\lambda^{-d} \subseteq \lambda} (-1)^{ht(\delta/\delta^{-d})} (-1)^{ht(\lambda/\lambda^{-d})} \chi_{S_{r-d}}^{\delta^{-d}}(\gamma_\alpha) \chi_{S_{r-d}}^{\lambda^{-d}}(\gamma_\alpha) \\ &= \sum_{d|k} \sum_{\delta^{-d} \subseteq \delta} \sum_{\lambda^{-d} \subseteq \lambda} (-1)^{ht(\delta/\delta^{-d})} (-1)^{ht(\lambda/\lambda^{-d})} \sum_{\alpha \vdash (r-d)} \frac{1}{z_\alpha} \chi_{S_{r-d}}^{\delta^{-d}}(\gamma_\alpha) \chi_{S_{r-d}}^{\lambda^{-d}}(\gamma_\alpha). \end{aligned}$$

By orthogonality of S_{r-d} characters, the inner-most sum is equal to $\langle \chi_{S_{r-d}}^{\delta^{-d}}, \chi_{S_{r-d}}^{\lambda^{-d}} \rangle = \delta_{\delta^{-d}, \lambda^{-d}}$, where $\delta_{\delta^{-d}, \lambda^{-d}}$ is a Kronecker delta. Thus, we must have $\lambda^{-d} = \delta^{-d}$ to get a nonzero inner product, and the proposition is proved. ■

The following theorem is our analogue of the Murnaghan-Nakayama rule. For a partition λ , recall the definition of λ^* in (1.4.2). Since we will be using recursion on the n , we will sometimes add the subscript n to the definition of d_μ from (2.1.2)

$$d_{\mu,n} = d_\mu = \gamma_\mu \otimes E_1^{\otimes n - |\mu|}, \quad \text{where } \mu \models n.$$

THEOREM 4.2.2. *Let $\lambda \vdash r$ with $|\lambda^*| \leq n$. Let μ be a composition with $0 \leq |\mu| \leq n$, and let $\bar{\mu}$ be the composition obtained from μ by removing a part of size k . Then*

$$\chi_{P_n(r)}^\lambda(d_{\mu,n}) = \sum_{d|k} \sum_{\substack{\delta = (\lambda^{-d}) + d \\ |\delta^*| \leq n-k}} (-1)^{ht(\lambda/\lambda^{-d})} (-1)^{ht(\delta/\lambda^{-d})} \chi_{P_{n-k}(r)}^\delta(d_{\bar{\mu},n-k}),$$

where the outer sum is over all divisors d of k and the inner sum is over all $\delta \vdash r$ with $|\delta^*| \leq n - k$ such that δ is obtained from λ by removing a border strip of size d to obtain the partition λ^{-d} and then adding a border strip of size d to λ^{-d} to obtain $\delta = (\lambda^{-d}) + d$.

Proof. Since $\bar{\mu} \models (n-k)$, our Frobenius formula (3.2.3) gives the following identity in $R(S_r)$:

$$r^{(n-k)-|\bar{\mu}|} f_{\bar{\mu}} = \sum_{\substack{\delta \vdash r \\ |\delta^*| \leq n-k}} \chi_{P_{n-k}(r)}^\delta(d_{\bar{\mu}, n-k}) \chi_{S_r}^\delta.$$

Using Proposition 4.2.1 and Corollary 3.2.5, we have

$$\begin{aligned} \chi_{P_n(r)}^\lambda(\gamma_\mu) &= \langle r^{n-|\mu|} f_\mu, \chi_{S_r}^\lambda \rangle \\ &= \langle r^{(n-k)-|\bar{\mu}|} f_{\bar{\mu}} f_k, \chi_{S_r}^\lambda \rangle \\ &= \sum_{\substack{\delta \vdash r \\ |\delta^*| \leq n-k}} \chi_{P_{n-k}(r)}^\delta(d_{\bar{\mu}, n-k}) \langle \chi_{S_r}^\delta f_k, \chi_{S_r}^\lambda \rangle \\ &= \sum_{\substack{\delta \vdash r \\ |\delta^*| \leq n-k}} \chi_{P_{n-k}(r)}^\delta(d_{\bar{\mu}, n-k}) \sum_{d|k} \sum_{\substack{\lambda-d \subseteq \lambda \\ \lambda-d \subseteq \delta}} (-1)^{ht(\lambda/\lambda^{-d})} (-1)^{ht(\delta/\lambda^{-d})} \\ &= \sum_{d|k} \sum_{\substack{\delta=(\lambda-d)+d \\ |\delta^*| \leq n-k}} (-1)^{ht(\lambda/\lambda^{-d})} (-1)^{ht(\delta/\lambda^{-d})} \chi_{P_{n-k}(r)}^\delta(d_{\bar{\mu}, n-k}). \end{aligned}$$

The last equality follows from changing the order of summation and observing that the inner product $\langle \chi_{S_r}^\delta f_k, \chi_{S_r}^\lambda \rangle$ is zero unless δ can be obtained from λ by removing and then adding back a border strip of size d . ■

COROLLARY 4.2.3. *If $\lambda \vdash r$ with $|\lambda^*| \leq n$ and μ is a composition with $0 \leq |\mu| \leq n$, then*

- (a) $\chi_{P_n(r)}^\lambda(d_\mu) = \begin{cases} r^{n-|\mu|} \chi_{P_{|\mu|}(r)}^\lambda(\gamma_\mu), & \text{if } |\mu| \geq |\lambda^*|, \\ 0, & \text{if } |\mu| < |\lambda^*|. \end{cases}$
- (b) *if $|\mu| = |\lambda^*| = n$, then $\chi_{P_n(r)}^\lambda(d_\mu) = \chi_{S_n}^{\lambda^*}(\gamma_\mu)$.*
- (c) *If $r \geq 2n$ and $|\mu| = n$, then $\chi_{P_n(r)}^\lambda(d_\mu)$ is independent of r .*

Proof. (a) In $P_{|\mu|}(r)$, we have $d_\mu = \gamma_\mu$. If $|\mu| \geq |\lambda^*|$, then

$$\chi_{P_n(r)}^\lambda(d_\mu) = \langle r^{n-|\mu|} f_\mu, \chi_{S_r}^\lambda \rangle = r^{n-|\mu|} \langle f_\mu, \chi_{S_r}^\lambda \rangle = r^{n-|\mu|} \chi_{P_{|\mu|}(r)}^\lambda(\gamma_\mu).$$

where the first and the third equality follow from Corollary 3.2.5.

If $|\mu| < |\lambda^*|$, then we recursively apply Theorem 4.2.2 until we have removed all of the parts of μ . We are left with a sum of characters of the form

$$\chi_{P_{n-|\mu|}(r)}^\delta(E_1^{\otimes(n-|\mu|)}) = \langle r^{n-|\mu|}, \chi_{S_r}^\delta \rangle = \begin{cases} r^{n-|\mu|}, & \text{if } \delta = (r), \\ 0, & \text{otherwise,} \end{cases}$$

where the first equality comes from Corollary 3.2.5 and the second equality comes from the fact that the only character that is constant on all S_r conjugacy classes is the trivial character $\chi_{S_r}^{(r)}$ which is 1 on all conjugacy classes. The partitions δ have minimal $|\delta^*|$, if at each step in the recursion we remove a strip of size μ_i from δ^* and add it to the first row of δ . In this case, we will have removed $|\mu|$ boxes from λ^* to obtain δ^* . However, $|\lambda^*| > |\mu|$, so $|\delta^*| > 0$. In particular $\delta \neq (r)$, and so the above inner product is 0 for each summand.

For (b), note that in Theorem 4.2.2, if $\lambda \vdash n$, then when we remove and add a border strip of size $d|k$ from λ to obtain δ , we must have $|\delta^*| \leq n - k$. We are thus forced to remove a border strip of size k from λ^{-d} and place it at the end of the first row in δ . The strip we add to δ has weight 1, and so we only consider the weight of removing a border strip of size k from λ^{-d} . This is precisely the Murnaghan-Nakayama rule for S_n (Theorem 4.1.1).

For (c) we see that if $|\mu| = n$, then by (a) we have $\chi_{P_n(r)}^\lambda(d_\mu) = \chi_{P_n(r)}^\lambda(\gamma_\mu)$. If $\lambda \vdash r$ with $|\lambda^*| = n - \ell$, then since $r \geq 2n$, we have

$$\lambda_1 - \lambda_2 \geq \underbrace{r - (n - \ell)}_{\lambda_1} - \underbrace{(n - \ell)}_{\geq \lambda_2} = r - 2n + 2\ell \geq 2\ell,$$

so a border strip in λ that contains boxes from λ_1 and λ_2 has at least $2\ell + 2$ boxes.

Suppose that d divides k and that we remove a border strip of size d from λ and add it back to obtain δ , with $|\delta^*| \leq n - k$. Assume that the border strip that we remove contains boxes from both λ_1 and λ_2 . If $k \geq \ell$, then we must move at least $k - \ell$ boxes from λ^* up to the first row of δ . But then $d = (\# \text{boxes removed from } \lambda^*) + (\# \text{boxes removed from } \lambda_1) \geq (k - \ell) + (2\ell + 1) = k + \ell + 1 > k$, a contradiction since d divides k . On the other hand, if $\ell > k$, then $d \geq \lambda_1 - \lambda_2 \geq 2\ell > k$, also a contradiction. Thus, we can not remove a strip from λ that contains boxes from both λ_1 and λ_2 , and in particular, we remove strips from λ_1 and λ^* independently.

An entirely similar argument shows that we can never add back a strip of size d that lives in both δ_1 and δ^* . As $r \geq 2n$ varies, the set of possible λ^* remains constant, only the length of the first row varies. Moreover, when $r \geq 2n$, there is only one way to remove or add a strip of size d from or to the first row. ■

Now we use Corollary 4.2.3 to lift our character formula to $P_n(x)$.

THEOREM 4.2.4. *Let $\lambda \vdash r$ with $|\lambda^*| \leq n$. Let μ be a composition with $0 \leq |\mu| \leq n$, and let $\bar{\mu}$ be the composition obtained from μ by removing a part of size k . Then*

- (a) if $|\mu| < n$, then $\chi_{P_n(x)}^\lambda(d_\mu) = \begin{cases} x^{n-|\mu|} \chi_{P_{|\mu|}}^\lambda(\gamma_\mu), & \text{if } |\mu| \geq |\lambda^*|, \\ 0, & \text{if } |\mu| < |\lambda^*|. \end{cases}$
- (b) if $|\mu| = n$, then $d_\mu = \gamma_\mu$, and

$$\chi_{P_n(x)}^\lambda(\gamma_\mu) = \sum_{d|k} \sum_{\substack{\delta=(\lambda^{-d})+d \\ |\delta^*| \leq n-k}} (-1)^{ht(\lambda/\lambda^{-d})} (-1)^{ht(\delta/\lambda^{-d})} \chi_{P_{n-k}(x)}^\delta(\gamma_{\bar{\mu}}),$$

where the outer sum is over all divisors d of k and the inner sum is over all $\delta \vdash r$ with $|\delta^*| \leq n-k$ such that δ is obtained from λ by removing a border strip of size d to obtain the partition λ^{-d} and then adding a border strip of size d to λ^{-d} to obtain $\delta = (\lambda^{-d})+d$.

Proof. Define $f_{\lambda\mu}(x) = 0$ if $|\mu| < |\lambda^*|$, and define

$$f_{\lambda\mu}(x) = x^{n-|\mu|} \sum_{d|k} \sum_{\delta=(\lambda^{-d})+d} (-1)^{ht(\lambda/\lambda^{-d})} (-1)^{ht(\delta/\lambda^{-d})} \chi_{P_{|\mu|-k}(x)}^\delta(\gamma_{\bar{\mu}}),$$

if $|\mu| < |\lambda^*|$. Then, by Corollary 4.2.3, for $r \geq 2n$ we have $f_{\lambda\mu}(r) = \chi_{P_n(r)}^\lambda(d_\mu)$. For an infinite number of $r \in \mathbb{Z}$, we have

$$f_{\lambda\mu}(r) = \chi_{P_n(r)}^\lambda(d_\mu) = \chi_{P_n(x)}^\lambda(d_\mu)|_{x=r}.$$

Since both $f_{\lambda\mu}(x)$ and $\chi_{P_n(x)}^\lambda(d_\mu)$ are rational functions in x which agree at an infinite number of points, then by Proposition 1.5.2 they must be equal everywhere. ■

5. BASIC CONSTRUCTION

To construct irreducible representations of $P_n(x)$, Martin [Ma3] exploits the properties of $P_0(x) \subseteq P_1(x) \subseteq P_2(x) \subseteq \dots$ as a tower of semisimple algebras. In this section, we derive general results about the characters of algebras which have these basic properties. Other algebras of this form are the rook algebra ([Mu1,2], [So2]) and its Iwahori-Hecke algebra ([So1], [So3], [Pu]) and the “rook” versions of the Brauer and Birman-Murakami-Wenzl centralizer algebras. Frobenius formulas and Murnaghan-Nakayama rules for the rook monoid and its Iwahori-Hecke algebra are determined in [DHP].

5.1. Basic Construction

Let $A \subseteq B$ be an inclusion of finite-dimensional, semisimple algebras with 1 over $\mathbb{C}$. Let $0 \neq e \in B$, and assume that the following properties

hold:

- (a) $e^2 = e$,
 - (b) $ea = ae$, for all $a \in A$,
 - (c) $A \cong Ae = eBe$ via the map $a \mapsto ae$ for all $a \in A$;
- (5.1.1)

Property (c) allows us to define a map,

$$\varepsilon : B \rightarrow A \quad \text{determined by } ebe = \varepsilon(b)e \text{ for all } b \in B. \quad (5.1.2)$$

PROPOSITION 5.1.3. *Let B be a semisimple algebra with 1 and let $e \in B$ such that $e \neq 0$ and $e^2 = e$. View Be as a module for BeB by multiplication on the left and as a module for eBe by multiplication on the right. Then*

$$BeB \cong \text{End}_{eBe}(Be) \quad \text{and} \quad eBe \cong \text{End}_{BeB}(Be).$$

Proof. Left multiplication by BeB commutes (by associativity) with right multiplication by eBe , so we have $BeB \subseteq \text{End}_{eBe}(Be)$ and $eBe \subseteq \text{End}_{BeB}(Be)$. Suppose that $\psi \in \text{End}_{BeB}(Be)$. Then $\psi(e) \in Be$ so $\psi(e) = \psi(e)e$. Since $be = be1 \in BeB$, we have

$$\psi(be) = \psi(bee) = be\psi(e) = bee\psi(e)e,$$

and so $\psi(be) = be x$ where $x = e\psi(e)e \in eBe$. Thus $eBe \cong \text{End}_{BeB}(Be)$.

Now, double centralizer theory (for example [CR], §3D) tells us the following: if A is semisimple, V is an A -module by the representation ρ , and $C = \text{End}_A(V)$ is the centralizer of A on V , then $\rho(A) = \text{End}_C(V)$. In particular, if ρ is faithful, then $A \cong \text{End}_C(V)$. In our case, B is semisimple and thus the ideal BeB is semisimple, so we only need to show that BeB acts faithfully on Be . To this end, assume that $(beb')Be = 0$. Then $(beb')BeB = 0$, but this is the left regular representation of BeB which is faithful. Thus $beb' = 0$ proving that $BeB \cong \text{End}_{eBe}(Be)$. ■

Using (5.1.1) (c), we have $A \cong eBe$ via $a \mapsto ae$. Right multiplication by A makes Be an A -module. Proposition 5.1.3 tells us that

$$BeB \cong \text{End}_A(Be) \quad \text{and} \quad A \cong \text{End}_{BeB}(Be). \quad (5.1.4)$$

Let $\widehat{X}$ denote the index set for the irreducible representations of the algebra X . Double centralizer theory applied to (5.1.4) gives $\widehat{BeB} = \widehat{A}$. Since BeB is an ideal of B , and B is semisimple, there exists an ideal $C \subseteq B$ so that

$$B \cong BeB \oplus C \quad (5.1.5)$$

and thus

$$\widehat{B} = \widehat{A} \sqcup \widehat{C}, \quad (5.1.6)$$

where $\sqcup$ denotes a disjoint union.

LEMMA 5.1.7. *If χ is a character of BeB , then χ is completely determined by the values $\chi(ae)$ where $a \in A$.*

Proof. (cf., [HR] Lemma 2.8). If $b_1eb_2 \in BeB$, then from the trace property of χ and the map ε defined in (5.1.2), we have $\chi(b_1eb_2) = \chi(b_1eeb_2) = \chi(eb_2b_1e) = \chi(\varepsilon(b_2b_1)e)$. ■

LEMMA 5.1.8. *If Q is a set of minimal orthogonal idempotents for A , then $eQ = \{eq \mid q \in Q\}$ is a set of minimal orthogonal idempotents for B . Moreover, q is a minimal idempotent of A corresponding to $\lambda \in \widehat{A}$ if and only if eq is a minimal idempotent of B corresponding to $\lambda \in \widehat{A}$.*

Proof. The proof of the first statement uses (5.1.1) (c) and the map ε defined in (5.1.2). We have $(eq)^2 = e^2q^2 = eq$, and for $q \neq q' \in Q$, $eqeq' = eeqq' = 0$. If $b \in B$, then $(eq)b(eq) = q(ebe)q = q\varepsilon(b)eq = q\varepsilon(b)qe = \alpha qe = \alpha(eq)$, for some scalar α , so eq is minimal.

Let z_λ be a minimal central idempotent of A such that $qz_\lambda \neq 0$. Since $A \cong eBe$ via the isomorphism $a \mapsto ae$, we have $qz_\lambda e \neq 0$. By the fact that A and BeB are full centralizers on Be , their centers coincide, and thus there is a minimal central idempotent z_λ^* in BeB which acts by left multiplication on Be the same way that z_λ acts on the right. In particular, $0 \neq qz_\lambda e = qez_\lambda = z_\lambda^*qe$, which shows that qe corresponds to $\lambda \in \widehat{B}$. ■

PROPOSITION 5.1.9. *If $\lambda \in \widehat{B}$ and $a \in A$, then*

$$\chi_B^\lambda(ae) = \begin{cases} \chi_A^\lambda(a), & \text{if } \lambda \in \widehat{A}, \\ 0, & \text{if } \lambda \in \widehat{B} \setminus \widehat{A}. \end{cases}$$

Proof. (cf., [HR] Proposition 2.10). Let P be a complete set of minimal orthogonal idempotents of A . Then $1 = \sum_{p \in P} p$. For $a \in A$, we have

$$ae = \left(\sum_{p \in P} p \right) ae \left(\sum_{p' \in P} p' \right) = \sum_{p, p' \in P} paep' = \sum_{p, p' \in P} (ep)a(ep') \quad (5.1.10)$$

Using Lemma 5.1.8, extend eP to a complete set Q of minimal orthogonal idempotents of B . For $b \in B$ and $q \in Q$, let $b|_q$ denote the coefficient of q when b is expanded in terms of Q . Let Q^λ denote the elements of Q corresponding to the irreducible component of B indexed by λ (define P^λ

similarly for A). Using (5.1.10), we have

$$\chi_B^\lambda(ae) = \sum_{q \in Q^\lambda} (qaeq)|_q = \sum_{q \in Q^\lambda} \sum_{p, p' \in P} q(ep)a(ep')q|_q$$

Since $q, ep \in Q$, we have $q(ep) = (ep)q = \begin{cases} ep, & \text{if } ep = q, \\ 0, & \text{otherwise.} \end{cases}$ Moreover, $ep \in Q^\lambda$ if and only if $p \in P^\lambda$ by Lemma 5.1.8, so

$$\chi_B^\lambda(ae) = \sum_{p \in P^\lambda} (ep)a(ep)|_{ep} = \sum_{p \in P^\lambda} pap|_p = \chi_A^\lambda(a)$$

as desired. ■

Let Ξ_B denote the character table for B , that is the table which has rows indexed by $\widehat{B}$ and columns indexed by appropriate basis elements of B whose character values are sufficient to determine the characters of B . Proposition 5.1.8 tells us that we can order the rows and columns of Ξ_B so that it takes the form

$$\Xi_B = \begin{bmatrix} \Xi_A & \vdots & * \\ \cdots & & \cdots \\ 0 & \vdots & \Xi_C \end{bmatrix}, \quad (5.1.11)$$

where Ξ_C is the character table of C .

5.2. Application to the Partition Algebras

There is a natural inclusion

$$P_{n-1}(x) \subseteq P_n(x) \quad (5.2.1)$$

given by adding a horizontal edge at the right end of the partition diagram of $d \in P_{n-1}(x)$. Let $e_n = \frac{1}{x}E_n$. Then it can be readily checked by drawing diagrams that $e_n^2 = e_n$ and $e_na = ae_n$ for all $a \in P_{n-1}(x)$. Furthermore,

$$\begin{array}{ccc} P_{n-1}(x) & \rightarrow & e_n P_n(x) e_n \\ a & \mapsto & ae_n \end{array} \quad (5.2.2)$$

is an algebra isomorphism. Thus, $P_{n-1}(x) \subseteq P_n(x)$ satisfies our properties (5.1.1) of a basic construction.

For a partition diagram d , we follow Martin [Ma4] and define the propagating number, $\#(d)$ to be the number of distinct parts of d containing elements from both the top and bottom row of d . Then if d_1, d_2 are partition diagrams, we have

$$\#(d_1 d_2) \leq \min(\#(d_1), \#(d_2)).$$

The ideal $P_n(x)e_nP_n(x)$ is spanned by all the diagrams having a propagating number strictly less than n . Moreover, we have ([Ma3])

$$P_n(x) \cong P_n(x)e_nP_n(x) \oplus \mathbb{C}[S_n]. \quad (5.2.3)$$

It follows from (5.1.5) or directly from the definition (1.4.1) that the index set $\widehat{P}_n$ for the irreducible representations of $P_n(r)$ is a disjoint union

$$\widehat{P}_n = \widehat{P}_{n-1} \sqcup \widehat{S}_n. \quad (5.2.4)$$

From Lemma 5.1.7 we know that if $\lambda \in \widehat{P}_{n-1}$, then χ^λ is completely determined by the values $\chi^\lambda(ae_n)$ for $a \in P_{n-1}(x)$. Moreover, by Proposition 5.1.9, if $\lambda \in \widehat{P}_n$ and $a \in P_{n-1}(x)$, then

$$\chi_{P_n(x)}^\lambda(ae_n) = \begin{cases} \chi_{P_{n-1}(x)}^\lambda(a), & \text{if } |\lambda^*| < n, \\ 0, & \text{if } |\lambda^*| = n. \end{cases} \quad (5.2.5)$$

The character table $\Xi_{P_n(x)}$ has the form of (0.1). We see this in the examples in section 6.

6. CHARACTER TABLES

The following tables give the characters of $P_n(x)$ for $n = 2, 3, 4, 5$. The entries in the tables are $\chi_{P_n(x)}^\lambda(d_\mu)$. The rows of the tables are indexed by the irreducible representations by the partitions λ^* defined in (1.4.2), and the columns of the table are indexed by the partitions μ , which correspond to the standard element d_μ defined in (2.1.2). The tables for $P_{n-1}(x)$ is embedded in the table for $P_n(x)$ according to (0.1), and so to save space we only give the rightmost part, where $|\mu| = n$, of the tables for $n = 4$ and $n = 5$. The tables for $P_n(r), r \geq n$, are obtained by setting $x = r$.

TABLE 1.
Characters of $P_2(x)$

	$\emptyset$	(1)	(2)	(1 ²)
$\emptyset$	x^2	x	2	2
(1)	0	x	1	3
(2)	0	0	1	1
(1 ²)	0	0	-1	1

TABLE 2.
Characters of $P_3(x)$

	$\emptyset$	(1)	(2)	(1 ²)	(3)	(2, 1)	(1 ³)
$\emptyset$	x^3	x^2	$2x$	$2x$	2	3	5
(1)	0	x^2	x	$3x$	1	4	10
(2)	0	0	x	x	0	2	6
(1 ²)	0	0	$-x$	1	0	0	6
(3)	0	0	0	0	1	1	1
(2, 1)	0	0	0	0	-1	0	2
(1 ³)	0	0	0	0	1	-1	1

TABLE 3.
Characters of $P_4(x)$

	(4)	(3, 1)	(2 ²)	(2, 1 ²)	(1 ⁴)
$\emptyset$	3	3	7	7	15
(1)	1	4	5	13	37
(2)	1	1	7	9	31
(1 ²)	-1	1	-5	5	31
(3)	0	1	2	4	10
(2, 1)	0	-1	0	2	20
(1 ³)	0	1	-2	-2	10
(4)	1	1	1	1	1
(3, 1)	-1	0	-1	1	3
(2 ²)	0	-1	2	0	2
(2, 1 ²)	1	0	-1	-1	3
(1 ⁴)	-1	1	1	-1	1

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TABLE 4.
Characters of $P_5(x)$

	(5)	(4, 1)	(3, 2)	(3, 1 ²)	(2 ² , 1)	(2, 1 ³)	(1 ⁵)
$\emptyset$	2	4	5	7	12	20	52
(1)	1	5	5	13	19	47	151
(2)	0	2	3	7	16	42	160
(1 ²)	0	0	-1	7	0	32	160
(3)	0	1	3	3	11	21	75
(2, 1)	0	0	-2	0	2	22	150
(1 ³)	0	-1	1	3	-9	1	75
(4)	0	1	1	3	3	7	15
(3, 1)	0	-1	0	0	1	9	45
(2 ²)	0	0	-1	-3	2	2	30
(2, 1 ¹)	0	1	0	0	-3	-3	45
(1 ⁴)	0	-1	1	3	-1	-5	15
(5)	1	1	1	1	1	1	1
(4, 1)	-1	0	-1	1	0	2	4
(3, 2)	0	-1	1	-1	1	1	5
(3, 1 ²)	1	0	0	0	-2	0	6
(2 ² , 1)	0	1	-1	-1	1	-1	5
(2, 1 ³)	-1	0	1	1	0	-2	4
(1 ⁵)	1	-1	-1	1	1	-1	1

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